

Project 2: Exploratory Data Analysis (EDA)

DecodeLabs Industrial Training Kit, Data Analytics Internship | Prepared by Sanni David Semilore

1. Dataset Overview

This analysis was performed on the cleaned dataset from Project 1 (Dataset_for_Data_Analytics_CLEANED.xlsx), containing 1,200 order records across 14 columns, covering orders placed between January 2023 and June 2025.

2. Methodology

Descriptive statistics (mean, median, count) were calculated for the key numeric fields (Quantity, UnitPrice, ItemsInCart, TotalPrice) using Excel formulas. Trends were explored using PivotTables across Product, Date, and PaymentMethod. Outliers were identified using the Interquartile Range (IQR) method: values above $Q3 + 1.5 \times IQR$ were flagged as statistical outliers.

3. Basic Statistics

Metric	Quantity	UnitPrice	ItemsInCart	TotalPrice
Mean	2.95	356.41	5.49	1,053.97
Median	3	364.21	5	823.62
Count	1,200	1,200	1,200	1,200

4. Outlier Detection (IQR Method)

Metric	Value
Maximum single order value	3,456.40
Upper outlier threshold ($Q3 + 1.5 \times IQR$)	3,330.41
Orders flagged as outliers	8 (0.67% of dataset)

The 8 flagged orders represent a very small share of the dataset (0.67%) and are consistent with naturally occurring high-value orders (e.g. bulk purchases) rather than data entry errors. No corrective action was taken, as the values fall within a plausible business range.

5. Key Observations

- Sales are spread evenly across products and payment methods, with no single product or payment channel dominating the dataset.

- The year 2025 shows lower total sales than 2023 and 2024, but this reflects incomplete data (only Q1 and Q2 are present) rather than a genuine decline in sales.
- The Mean TotalPrice (1,053.97) is notably higher than the Median (823.62), indicating the distribution is right-skewed, meaning a smaller number of high-value orders are pulling the average upward.
- 8 orders (about 0.67% of the dataset) fall above the statistical outlier threshold of 3,330.41, but this is a small and expected amount, not indicative of a data quality issue.
- The average UnitPrice varies widely across products, and the average cart contains approximately 5–6 items, suggesting customers typically purchase in small multiples rather than single units.

6. Conclusion

The dataset shows a healthy, well-distributed sales pattern with no major red flags. The right-skewed TotalPrice distribution and small set of high-value outliers are typical of order-level retail data and do not require further cleaning. The main caveat for any downstream analysis is that 2025 data is incomplete and should not be directly compared year-over-year against 2023 or 2024 without adjusting for the partial year.